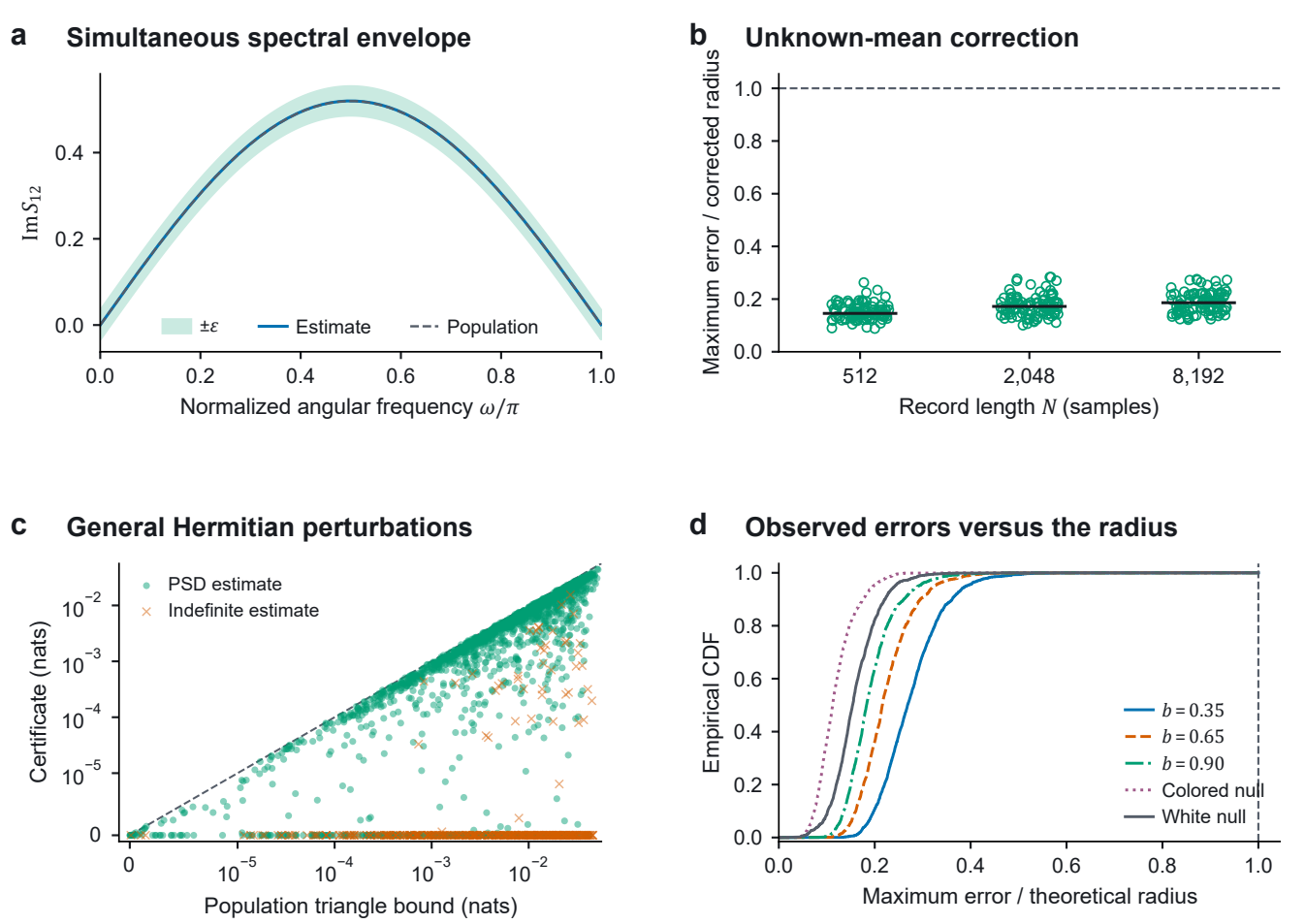

a
Simultaneous spectral envelope
$\mathrm{Im}\,S_{12}$
0.0
0.2
0.4
±ε
Estimate
Population
0.0
0.2
0.4
0.6
0.8
1.0
Normalized angular frequency ω/π
b
Unknown-mean correction
Maximum error / corrected radius
0.0
0.2
0.4
0.6
0.8
1.0
512
2,048
8,192
Record length N (samples)
c
General Hermitian perturbations
PSD estimate
Indefinite estimate
Certificate (nats)
10^{-2}
10^{-3}
10^{-4}
10^{-5}
0
0
10^{-5}
10^{-4}
10^{-3}
10^{-2}
Population triangle bound (nats)
d
Observed errors versus the radius
Empirical CDF
0.0
0.2
0.4
0.6
0.8
1.0
$b = 0.35$
$b = 0.65$
$b = 0.90$
Colored null
White null
0.0
0.2
0.4
0.6
0.8
1.0
Maximum error / theoretical radius